Surface Reduction and Self-Passivation in a Single-Step Ar/CH_4 Plasma treatment for Hybrid Cu Bonding

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***Abstract*—This study investigates Ar/CH_4 plasma pre-treatment on Cu surface as a dual-functional approach that removes Cu oxides via hydrogen radicals/ion-assisted reactions while forming a hydrogenated amorphous carbon (a-C:H) passivation layer. Surface and interfacial evolution were examined by XPS, TOF-SIMS, SEM, and TEM, along with an evaluation of the a-C:H thickness effect. XPS and TEM verify effective oxide removal and time-dependent growth of a-C:H, exhibiting an approximately linear thickness increase with plasma duration. TOF-SIMS reveals a hydrogen concentration gradient across the a-C:H layer. Consistent with this depth-dependent composition, chemical-state analysis indicates that shortening the plasma exposure reduces the a-C:H thickness such that the remaining film is predominantly sp^2-rich. After bonding at 260 °C, cross-sectional TEM shows an ~15 nm interfacial carbon layer under longer Ar/CH_4 plasma exposure, whereas direct Cu-to-Cu bonding is observed for the shortest treatment. While TEM indicates void-free interfaces irrespective of carbon thickness in the inspected regions, low-magnification SEM reveals voids that are primarily attributable to pre-existing Cu surface roughness prior to plasma treatment. These results demonstrate the potential of Ar/CH_4 plasma pre-treatment for enabling low-temperature hybrid Cu bonding through concurrent oxide removal and formation of a controllable a-C:H passivation layer.**

Keywords— Hybrid Cu bonding, Reduction of Cu oxides, Passivation with a-C:H, Concurrent reduction and passivation

I. Introduction

Advanced three-dimensional (3D) integration has become a central pathway for sustaining system-level scaling as conventional transistor scaling faces increasing limitations. In this context, hybrid Cu bonding has emerged as a leading interconnect solution because it enables fine-pitch, gap-less electrical and dielectric interfaces, which are essential for high-density stacking in memory-logic integration and heterogeneous systems. To fully realize these advantages in manufacturing, the bonding temperature must be lowered to mitigate thermal stress, warpage, and reliability degradation in temperature-sensitive device stacks [1, 2].

A key challenge in reducing the bonding temperature is the intrinsic requirement for interfacial atomic transport. Sufficient Cu-Cu contact formation typically relies on surface cleanliness and atomic diffusion processes, while excessive diffusion at elevated temperatures can induce reliability concerns in both DRAM and logic devices. In addition, native copper oxides (e.g., Cu_2O/CuO) form rapidly in ambient conditions and act as diffusion barriers that hinder metallic bonding, increase interfacial resistance, and promote non-uniform bonding. Therefore, surface preparation prior to bonding is a decisive step for low-temperature hybrid bonding.

In particular, plasma-based pre-bond treatments have been extensibely investigated because they can (i) remove copper oxides, (ii) activate Cu and dielectric surfaces, and (iii) tailor near-surface chemistry without introducing excessive thermal load. Conventional approaches often focus on oxide removal; however, the freshly reduced Cu surface is highly reactive and may re-oxidize before bonding, particularly during wafer handling and alignment. This has motivated the exploration of plasma chemistries that can simultaneously provide oxide reduction and passivation [3-6].

In this context, Ar/CH_4 plasma treatment has been proposed as an effective route for low-temperature Cu bonding through a dual mechanism: (1) reduction of Cu oxides via hydrogen radicals and ion-assisted reactions, and (2) passivation of the cleaned Cu surface via the formation of a hydrogenated amorphous carbon (a-C:H) layer. In this study, XPS and TEM confirm efficient oxide removal and the formation of a uniform carbon layer (on the order of several nanometers) consisting of mixed sp^2/sp^3 carbon clusters. In addition, TOF-SIMS depth profiles suggest a hydrogen concentration gradient in the a-C:H layer, where the near-surface region is relatively hydrogen-rich (often associated with sp^3-rich bonding environments) while the region closer to Cu is comparatively sp^2-rich. Although such a-C:H passivation can suppress re-oxidation and improve process stability, its presence introduces a fundamental concern for hybrid bonding: an interfacial carbon layer must not remain after bonding, because residual carbon can obstruct metallic contact, degrade electrical performance, and compromise long-term reliability. Accordingly, this work additionally investigates the thickness effect of the a-C:H layer, examining how plasma exposure conditions modulate carbon thickness and sp^2/sp^3 bonding characteristics in relation to bonding behavior.

II. Method

An 8-inch notch-type Si wafer with a 700 nm SiO_2 layer was used as the substrate. Ti (50 nm) and Cu (1 μm) films were deposited by DC sputtering. The Cu surface was then treated using an ICP-based Ar/CH_4 plasma at an ICP power of 2000 W, a pressure of 4 mTorr, and gas flow rates of Ar 30 sccm and CH_4 70 sccm. This DOE-derived condition yields an approximately 7 nm hydrogenated amorphous carbon (a-C:H) layer on Cu. To examine the thickness-dependent Cu diffusivity and bonding behavior, the plasma exposure time alone was varied across four levels based on the DOE-derived recipe, while all other plasma parameters were held constant.

Cu-to-Cu bonding was performed at 260 °C. Surface chemistry and carbon-layer characteristics were analyzed by XPS and TOF-SIMS, while microstructural characterization of both the plasma-treated surface and the bonded interface was conducted using SEM and TEM. Bond integrity and electrical continuity were evaluated by shear testing and daisy-chain measurements, respectively.

III. Results and Discussions

Surface chemical modification by Ar/CH_4 plasma treatment

The Cu LMM Auger spectra (Fig. 1) show that Cu_2O is effectively removed after Ar/CH_4 plasma treatment. Importantly, even the shortest exposure (15 s) yields a pronounced change in the Cu LMM line shape, consistent with substantial oxide reduction. As the plasma duration increases, the Cu-related Auger features progressively weaken due to attenuation by the increasingly thick a-C:H overlayer. Consequently, after 60 s of Ar/CH_4 plasma treatment, the Cu LMM signal becomes indistinguishable within the detection limit.

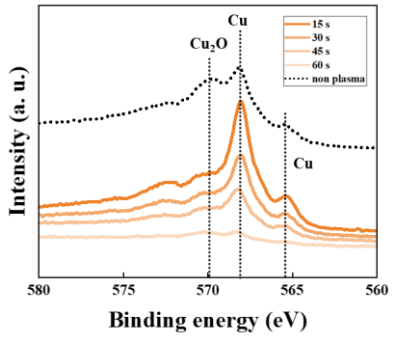


Fig. 1. Evolution of Cu LMM Auger spectra with plasma treatment time

Figure 2 shows that a uniform carbon layer is formed on the plasma-treated Cu surface. Here, P15 and P60 denote Ar/CH_4 plasma treatment times of 15 and 60 s, respectively. In conjunction with the Cu LMM Auger spectra, the P15 condition indicates that, despite the short plasma exposure, Cu oxides are effectively removed while a conformal a-C:H overlayer with a thickness of approximately 2.5 nm is simultaneously formed.

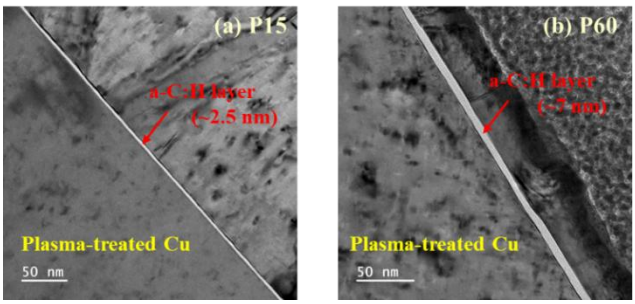


Fig. 2. Cross-sectional TEM images of the a-C:H layer on Cu after Ar/CH_4 plasma treatment (P15 vs. P60)

Bonding Interface Characterization by SEM

The bonding results show a clear dependence on the a-C:H thickness. For the P60 condition, a carbon interlayer remains observable at the bonded interface. In contrast, the P15 sample exhibits direct Cu–Cu bonding without a discernible carbon layer, as shown in Fig. 3. This observation suggests that, under the P15 condition, carbon species can diffuse into the Cu region at 260 °C, via interstitial pathways and grain-boundary diffusion. It is noted that the dark features observed along the P15 interface correspond to voids, which are attributed to the as-received Cu surface roughness rather than the presence of a carbon interlayer.

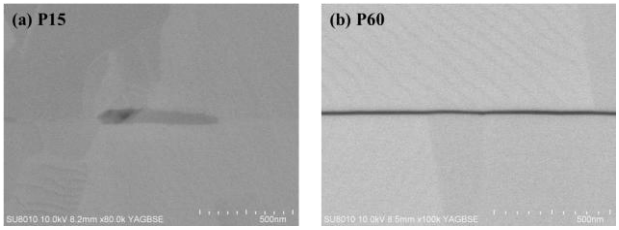


Fig. 3. Cross-sectional SEM of the bonded interface: P15 vs. P60

IV. Summary

In this section, This study examined Ar/CH_4 ICP plasma pre-treatment for low-temperature Cu hybrid bonding by varying only the plasma exposure time (four levels) under a DOE-derived recipe (2000 W, 4 mTorr, Ar 30 sccm, CH_4 70 sccm). Cu LMM confirmed effective oxide removal even at the shortest condition (P15), while cross-sectional TEM showed formation of a conformal a-C:H overlayer (~2.5 nm for P15) that thickened with increasing treatment time and progressively attenuated the Cu LMM signal (undetectable for P60). Bonding at 260 °C exhibited a thickness-dependent interface: P60 retained an interfacial carbon layer, whereas P15 showed direct Cu–Cu bonding without a discernible carbon interlayer; void-like dark features observed in P15 were attributed to the as-received Cu surface roughness rather than carbon residues. Collectively, these results highlight the potential of Ar/CH_4 plasma pre-treatment as a practical pathway toward low-temperature hybrid Cu bonding..

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